

Dynamic Guideline: WHO-Based Dengue - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest WHO clinical guideline for dengue-relevant management

- **Guideline Title:** *WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever*
- **Publication Date:** **4 July 2025**; WHO announced the guidance on **10 July 2025** as its first global guideline covering all four diseases together [WHO publication page](#), [WHO news](#).
- **Target Population:** Patients with **suspected or confirmed arboviral disease**, including dengue, managed in community care, primary care, emergency departments, inpatient wards, and higher-acuity hospital settings [WHO news](#).
- **Target Users:** Health-care providers, emergency and inpatient clinicians, public health programmes, and health-system planners caring for suspected or confirmed dengue and other arboviral infections [NCBI Bookshelf](#).
- **Scope and Objectives:** Supportive and symptomatic management of non-severe and severe arboviral infection; oral and intravenous fluid management; monitoring strategies to guide fluid administration; and avoidance of ineffective or harmful adjunctive therapies [NCBI Bookshelf](#).

Most recent WHO guideline-process document identified

WHO also published “**Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)**” in December 2025. This is a **guideline-development process evidence brief**, not a dengue clinical guideline. It is relevant methodologically because it describes how youth input can be incorporated into WHO guideline processes through consultations and youth-led convening, but no source links it directly to the 2025 arboviral guideline [WHO Western Pacific publication page](#), [WHO launch webinar page](#).

WHO / WHO-linked guideline corpus synthesized: exactly 5 documents

#	Document	Date	Dengue Relevance	Role in This Report
1	<i>WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever</i>	4 July 2025	Direct clinical management guideline for dengue and other arboviruses	Primary latest WHO clinical guideline WHO publication page

#	Document	Date	Dengue Relevance	Role in This Report
2	<i>Laboratory testing for dengue virus: interim guidance</i>	April 2025	Dengue diagnostics and laboratory-testing algorithm	Primary diagnostic guidance WHO publication page
3	<i>Dengue: Guidelines for diagnosis, treatment, prevention and control</i>	2009	Dengue-specific diagnosis, treatment, prevention, control	Historical WHO dengue foundation guideline PAHO-hosted WHO PDF
4	<i>PAHO Guidelines for the Clinical Diagnosis and Treatment of Dengue, Chikungunya, and Zika</i>	2022	Regional WHO/PAHO evidence-based clinical diagnosis and treatment guidance	GRADE-based regional clinical guideline informing WHO methods NCBI Bookshelf
5	<i>Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)</i>	9–11 December 2025	Not dengue-specific; guideline-process methodology	Included because it is the most recent WHO guideline-process document identified and addresses inclusive guideline development WHO Western Pacific publication page

1.2 Key Recommendations

GRADE interpretation used in this report

- **Quality of evidence:** High, Moderate, Low, Very Low.
- **Strength of recommendation:** Strong or Conditional.
- Where WHO explicitly reported certainty and recommendation strength, that rating is used. Where a source did not formally GRADE-rate a statement, this report provides a transparent **report-applied GRADE appraisal** based on study design, consistency, directness, and clinical plausibility.

Recommendation	Clinical Action	GRADE Evidence Quality	Strength
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Recommendation	Clinical Action	GRADE Evidence Quality	Strength
Suspect dengue in patients with acute febrile illness compatible with dengue and epidemiologic exposure; do not rely on clinical diagnosis alone when laboratory confirmation is available.	Use epidemiology, symptoms, warning signs, and laboratory testing by illness day. WHO states clinical diagnosis alone is insufficient because symptoms overlap with other diseases WHO publication page .	Moderate	Strong
For illness days 0–7 , use dengue RNA NAAT/RT-PCR and/or NS1 antigen testing when available.	CDC states dengue RNA is usually detectable by molecular tests during days 0–7, and NS1 antigen is detectable in the acute phase CDC testing guidance .	Moderate	Strong
After day 7, interpret IgM positivity as evidence of recent/probable dengue infection, with attention to cross-reactivity and local epidemiology.	WHO interim diagnostic guidance emphasizes test selection by days post symptom onset and notes that no universal algorithm fits all settings WHO publication page .	Moderate	Conditional
Classify patients into dengue without warning signs, dengue with warning signs, or severe dengue; monitor closely around defervescence.	Warning signs include abdominal pain, persistent vomiting, mucosal bleeding, lethargy/restlessness, hepatomegaly, fluid accumulation, and rising hematocrit with falling platelets NCBI Bookshelf , StatPearls .	Low	Strong
Treat dengue supportively; no specific antiviral therapy is currently recommended for routine care.	CDC states there are no specific medicines or antiviral agents for dengue; care is supportive CDC clinical care .	Moderate	Strong
Use acetaminophen/paracetamol for fever and pain; avoid aspirin and NSAIDs because of bleeding risk.	CDC recommends acetaminophen and avoidance of aspirin/NSAIDs such as ibuprofen CDC clinical care .	Moderate	Strong
Hospitalize patients with warning signs or severe dengue.	CDC advises hospital management for warning signs or severe dengue CDC clinical care . PAHO algorithms also list hospitalization criteria including warning signs and severe dengue PAHO algorithm PDF .	Low	Strong

Recommendation	Clinical Action	GRADE Evidence Quality	Strength
For severe suspected or confirmed arboviral disease requiring IV fluids, use crystalloid rather than colloid fluid.	WHO suggests crystalloids over colloids for severe arboviral disease requiring IV fluids ReliefWeb WHO guideline summary .	Low	Conditional
Use capillary refill time to guide IV fluid management in severe arboviral disease.	WHO recommends capillary refill time to guide IV fluids ReliefWeb WHO guideline summary .	Low	Strong
Use lactate measurement in addition to standard care to guide IV fluid management in severe arboviral disease when available.	WHO recommends lactate plus standard care ReliefWeb WHO guideline summary .	Moderate	Strong
Consider passive leg raise testing in shock when uncertain whether further IV fluid is warranted.	WHO suggests passive leg raise testing in shock when additional fluids are uncertain ReliefWeb WHO guideline summary .	Low	Conditional
Avoid corticosteroids in acute non-severe suspected or confirmed arboviral disease.	WHO suggests against corticosteroids for non-severe arboviral disease ReliefWeb WHO guideline summary .	Low	Conditional
Avoid systemic corticosteroids in severe suspected or confirmed arboviral disease, except for separate indications.	WHO suggests against systemic corticosteroids in severe arboviral disease ReliefWeb WHO guideline summary . CDC also states corticosteroids have no demonstrated dengue benefit and may be harmful except for autoimmune-related complications CDC clinical care .	Very Low	Conditional
Avoid immunoglobulins for suspected or confirmed severe arboviral disease.	WHO suggests against immunoglobulins ReliefWeb WHO guideline summary .	Very Low	Conditional
Avoid prophylactic platelet transfusion in severe suspected or confirmed arboviral disease with platelet count <50,000/ μ L and no active bleeding.	WHO suggests against prophylactic platelet transfusion without active bleeding ReliefWeb WHO guideline summary . CDC states prophylactic platelet transfusions are not beneficial and may contribute to fluid overload CDC clinical care .	Low	Conditional

1.3 Guideline Development Methodology

Evidence Review Process

- WHO's 2025 arboviral guideline used a GRADE-based approach for evidence synthesis and recommendation development, drawing on systematic review processes and evidence profiles [NCBI Bookshelf](#).
- The 2022 PAHO guideline process, which informed the evidence base for dengue, chikungunya, and Zika, used electronic database searches, manual searches, evidence profiles, and the **Grading of Recommendations Assessment, Development and Evaluation (GRADE)** approach [NCBI Bookshelf](#).
- WHO's 2025 laboratory-testing guidance synthesizes current evidence on dengue diagnostics and emphasizes that test choice should be adapted to days after symptom onset, sample type, local epidemiology, prior flavivirus exposure, vaccination status, and co-circulating pathogens [WHO publication page](#).

Consensus Method

- Recommendations were agreed by a panel of arbovirus experts using WHO guideline-development methods [NCBI Bookshelf](#).
- For recommendation strength, WHO/GRADE distinguishes:
 - **Strong recommendations:** most patients should receive the recommended course.
 - **Conditional recommendations:** patient values, setting, feasibility, resources, and uncertainty may lead to different choices [Guideline Central](#).

Conflict of Interest Management

- The supplied sources do not provide detailed conflict-of-interest declarations for the 2025 arboviral guideline panel.
- The older WHO dengue guideline reported that declarations of interest were obtained, and no conflicting interests were declared [PAHO-hosted WHO PDF](#).

Equity, Acceptability, and Guideline-Process Inclusivity

- WHO's 2025 evidence brief on meaningful youth engagement states that youth input can strengthen dialogue, enrich policy development, and help ensure global health guidance reflects the realities of affected populations [WHO launch webinar page](#).
- No identified WHO source states that this youth-engagement model was applied to the 2025 arboviral clinical management guideline.

II. Evidence Summary After WHO Latest Guideline Release (WHO guidelines for clinical management of arboviral diseases: dengue, chikungunya, Zika and yellow fever) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence was reviewed from the period **4 July 2025 to 29 April 2026**, focusing on dengue diagnosis, acute treatment, severe-dengue management, fluid therapy, vaccines, antivirals, and ongoing trials.

Databases and Sources Searched

- WHO official publication pages and WHO news updates.
- CDC dengue clinical care and diagnostic guidance.
- PAHO dengue clinical management algorithms.
- NCBI Bookshelf guideline text.
- PubMed-indexed and open-access clinical reviews.
- ClinicalTrials.gov records for dengue antiviral, vaccine, and severe-dengue studies.
- Vaccine trial updates from Takeda and CIDRAP summaries.

Inclusion Criteria

Included evidence addressed at least one of the following:

1. Dengue diagnosis or diagnostic accuracy.
2. Acute dengue treatment or supportive management.
3. Severe dengue, shock, bleeding, or fluid management.
4. Dengue vaccine efficacy/safety evidence relevant to prevention.
5. Registered or updated clinical trials likely to affect future dengue management.

Exclusion Criteria

Excluded evidence if it:

- Did not address dengue or arboviral disease.
- Was unrelated to diagnosis, treatment, management, prevention, or guideline methodology.
- Had no accessible study details in the supplied evidence summaries.
- Was superseded by the July 2025 WHO clinical guideline unless needed for context.

2.2 Key New Studies

Study / Evidence Source	Design	Key Findings	GRADE Quality	Practice Impact
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Study / Evidence Source	Design	Key Findings	GRADE Quality	Practice Impact
TAK-003/QDENGALong-term TIDES follow-up, reported November 2025	Phase 3 vaccine trial follow-up	Two-dose TAK-003/QDENGALong-term TIDES follow-up, reported November 2025 showed sustained protection. CIDRAP reported 61.2% efficacy against virologically confirmed dengue and 84.1% efficacy against dengue-associated hospitalization after 4.5 years; hospitalization efficacy increased to 90.6% two years after booster. No new safety signals were reported CIDRAP, Takeda.	Moderate	Supports vaccination as prevention where authorized and recommended; does not change acute dengue treatment.
Dengue Fever Vaccines: Progress and Challenges	2026 narrative review	Described QDENGALong-term TIDES follow-up, reported November 2025 pivotal phase 3 evidence, two-dose schedule at 90-day interval, efficacy against virologically confirmed dengue, and outcomes by serotype and serostatus Annual Reviews.	Low	Reinforces prevention importance; no change to severe-dengue fluid management.
Adaptive Dengue Antiviral Platform Trial, NCT06551844	Ongoing adaptive antiviral platform trial	Enrolls patients with clinical dengue, fever <48 hours, positive NS1 rapid test, and excludes severe dengue at baseline. No results available in supplied record ClinicalTrials.gov NCT06551844.	Very Low	No current practice change; important research priority.

Study / Evidence Source	Design	Key Findings	GRADE Quality	Practice Impact
Innovative Multidisciplinary Strategies for Combating Severe Dengue, NCT06751836	Observational severe-dengue/pathogenesis study	Enrolling dengue without warning signs, dengue with warning signs, severe dengue, and controls; uses multi-omics to study immune responses and severity prediction. Estimated enrollment 1,400; no results yet ClinicalTrials.gov NCT06751836 .	Very Low	No immediate clinical recommendation change; may improve severity prediction in future.
Dengue tetravalent vaccine safety study in adults aged 45–60 and >60–79 years, NCT06579755	Vaccine safety study	Evaluates solicited systemic and local adverse events after Takeda dengue tetravalent vaccine; last updated 22 April 2026 ClinicalTrials.gov NCT06579755 .	Very Low	No current acute-care impact; may inform vaccine use in older adults.
Dengue tetravalent vaccine in healthy participants in Japan, NCT06741683	Phase 2/3 randomized, double-blind/placebo-controlled and open-label vaccine trial	Evaluates immunogenicity and safety of two subcutaneous doses in adults, adolescents, and children in Japan; no results posted in supplied record ClinicalTrials.gov NCT06741683 .	Very Low	No current practice change.
CDC updated clinical care guidance	Authoritative public health guidance	Reaffirms no specific antiviral treatment; supportive care; hydration; acetaminophen; avoidance of aspirin/NSAIDs; hospital management for warning signs or severe dengue; no benefit from prophylactic platelet transfusion CDC clinical care .	Moderate	Consistent with WHO 2025; strengthens implementation message.

Study / Evidence Source	Design	Key Findings	GRADE Quality	Practice Impact
CDC diagnostic guidance	Authoritative public health guidance	Recommends testing by illness phase: RNA molecular tests and NS1 antigen during days 0–7; reduced usefulness after day 7; dengue is nationally notifiable in the United States CDC testing guidance .	Moderate	Consistent with WHO laboratory guidance; no conflicting recommendation.

2.3 Evidence Gaps and Future Research Needs

- 1. No dengue-specific antiviral has proven clinical benefit for routine care.**
Current guidance remains supportive; the adaptive antiviral platform trial is important but has no practice-changing results yet [ClinicalTrials.gov NCT06551844](#).
- 2. Limited new post-July 2025 evidence on severe-dengue fluid strategy.**
No new randomized trial or systematic review was identified that changes WHO’s recommendations on crystalloid fluids, capillary refill, lactate monitoring, or passive leg raise testing.
- 3. Need better severity prediction tools.**
Ongoing severe-dengue multi-omics research may help identify biomarkers and pathways associated with progression to severe disease [ClinicalTrials.gov NCT06751836](#).
- 4. Diagnostic accuracy varies by illness day, primary vs secondary infection, and local flavivirus context.**
WHO states that no universal diagnostic algorithm exists and that local adaptation is required [WHO publication page](#).
- 5. Vaccine implementation questions remain.**
New vaccine evidence supports prevention, but policy decisions still need to account for serostatus, age, local dengue burden, serotype circulation, product authorization, and safety monitoring [CIDRAP](#), [Takeda](#).
- 6. Guideline equity and engagement methods need broader application.**
WHO’s youth-engagement evidence brief offers a model for inclusive guideline development, but no source confirms use of that approach in dengue guideline development [WHO Western Pacific publication page](#).

2.4 Updated Recommendations Based on New Evidence

Original WHO-Based Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation	GRADE Quality	Strength
Manage dengue supportively; no routine antiviral therapy.	No positive dengue antiviral trial results identified; antiviral platform trial ongoing ClinicalTrials.gov NCT06551844 .	No change. Provide supportive care, risk stratification, hydration, monitoring, and escalation for warning signs/severe dengue.	Moderate	Strong
Use illness-day-based laboratory testing.	CDC guidance remains consistent with WHO: RNA/NS1 during days 0–7; serology later CDC testing guidance .	No change. Use NAAT/RT-PCR and/or NS1 early; use IgM later with awareness of cross-reactivity.	Moderate	Strong
Use crystalloids rather than colloids when IV fluids are required for severe arboviral disease.	No new post-guideline fluid trial evidence identified that changes this.	No change. Use isotonic crystalloid as first-line IV fluid; reassess frequently.	Low	Conditional
Use capillary refill time and lactate to guide IV fluid management.	No contradictory evidence identified.	No change. Use capillary refill routinely; add lactate where available.	Low for capillary refill; Moderate for lactate	Strong
Consider passive leg raise when in shock and uncertain about more fluids.	No new evidence changes this conditional recommendation.	No change. Use as an adjunct to avoid under- or over-resuscitation when feasible.	Low	Conditional
Avoid corticosteroids in non-severe and severe dengue/arboviral disease unless another indication exists.	CDC continues to state no demonstrated dengue benefit and potential harm, except selected autoimmune complications CDC clinical care .	No change. Do not use corticosteroids for dengue itself.	Low to Very Low	Conditional

Original WHO-Based Recommendation	New Evidence Impact Through 29 April 2026	Updated Recommendation	GRADE Quality	Strength
Avoid prophylactic platelet transfusion without active bleeding.	CDC reiterates prophylactic platelet transfusions are not beneficial and may contribute to fluid overload CDC clinical care .	No change. Reserve transfusion for clinically significant bleeding or other accepted indications.	Low	Conditional
Consider dengue vaccination as a prevention strategy where recommended by national policy.	Long-term TAK-003/QDenga evidence supports sustained protection and reduced hospitalization; no new safety signals reported CIDRAP , Takeda .	Implementation-sensitive update. Vaccination can be incorporated into prevention programmes where authorized and aligned with WHO/national policy, but it does not alter acute dengue treatment.	Moderate	Conditional

Clinical Implementation Pathway

A. Initial Assessment

1. Assess epidemiologic risk: residence in or travel to dengue transmission area within the prior 14 days.
2. Identify acute febrile illness and compatible symptoms: headache, retro-orbital pain, myalgia, arthralgia, rash, nausea/vomiting, petechiae, or bleeding.
3. Evaluate for warning signs:
 - Abdominal pain or tenderness.
 - Persistent vomiting.
 - Clinical fluid accumulation.
 - Mucosal bleeding.
 - Lethargy or restlessness.
 - Hepatomegaly.
 - Rising hematocrit with falling platelets [NCBI Bookshelf](#), [StatPearls](#).

B. Diagnostic Testing

Illness Phase	Preferred Testing	Interpretation
Days 0–7	Dengue RNA NAAT/RT-PCR and/or NS1 antigen	Best for acute confirmation CDC testing guidance
>7 days	IgM serology, with or without paired serology depending on setting	Suggests recent/probable dengue; interpret with cross-reactivity and local flavivirus context WHO publication page

Illness Phase	Preferred Testing	Interpretation
Pregnancy with possible Zika exposure	Test for Zika by molecular methods in addition to dengue	Recommended by CDC because of overlapping arboviral syndromes CDC testing guidance

C. Outpatient Management for Dengue Without Warning Signs

- Encourage oral fluids.
- Use acetaminophen/paracetamol for fever and pain.
- Avoid aspirin, ibuprofen, and other NSAIDs.
- Provide strict return precautions, especially around defervescence.
- Advise mosquito-bite prevention while febrile to reduce transmission [CDC clinical care](#).

D. Inpatient Management for Warning Signs or Severe Dengue

- Admit patients with warning signs or severe dengue.
- Monitor vital signs, perfusion, urine output, mental status, hematocrit, platelets, and bleeding.
- Use isotonic crystalloid fluids when IV fluids are required.
- Reassess frequently to avoid both shock and fluid overload.
- Use capillary refill time and lactate, where available, to guide resuscitation.
- Avoid prophylactic platelet transfusion without active bleeding.
- Escalate to ICU-level care for shock, respiratory distress, severe bleeding, severe organ involvement, or need for close hemodynamic monitoring [CDC clinical care](#), [ReliefWeb WHO guideline summary](#).

Overall Conclusion

As of **29 April 2026**, the latest WHO dengue-relevant clinical guideline remains the **4 July 2025 WHO guideline for clinical management of arboviral diseases**, covering dengue, chikungunya, Zika, and yellow fever. The core dengue management approach remains unchanged: **early recognition, illness-day-appropriate testing, severity classification, supportive care, careful fluid management for severe disease, and avoidance of corticosteroids, immunoglobulins, and prophylactic platelet transfusion without bleeding**.

Post-guideline evidence through **29 April 2026** does **not** identify a new dengue-specific antiviral or a new severe-dengue fluid-management trial that changes WHO recommendations. Vaccine evidence, especially for TAK-003/QDENGGA, continues to strengthen prevention policy discussions, but it does not alter acute dengue diagnosis or treatment.